HOMEWORK:
Pooled OLS, Fixed Effects, and Random Effects

INSTRUCTIONS:

- This homework can be done individually or by groups of **2 or 3** students.
- You have to submit two files:
  1. An answers sheet. This file contains answers to the questions and should be in pdf format. Answers are a mix of typed text, equations, and estimation results. For estimation results, you should write in the answers sheet the command and create a table summarizing them.
  2. A do or R file that contains only the code used to answer to questions (use comments to help understanding the structure). This file should be ready to use to replicate the results presented in the answers sheet.
- **Important rule**: Any question for which no detailed comment on the estimation or test results is given will have a zero score, even if the code in R or Stata is correct.
- Submission due deadline: **Friday, March 13th at 12pm**. Late submissions will not be considered for grading.
- Submission procedure:
  - You must indicate with whom you submit the homework.
  - Answers have to be submitted on the course webpage, using the appropriate link, by one member of the group only.
- This homework counts for 40% in the final grade.

Exercice 1. Theory. (**20 points**)

Consider the regression equation

$$y_{it} = \alpha_i + \beta x_{it} + u_{it}, \text{ with } (u_{it}|X) \sim \text{IID}\left(0, \sigma^2\right) \tag{1}$$

where y_{it} denotes the outcome of interest, α_i, an individual-specific effect, x_{it}, an explanatory variable with associated impact measured by β, and u_{it}, the usual error terms which, conditionally on X, i.e. all values of x_{it}, are independently and identically distributed with mean 0 and variance σ^2. Non-balance in panel data set can be formalized by letting the individuals and periods from which observations are available represented by index sets, in that

$$\begin{aligned} I_t &= \quad \text{Index set of individuals observed in period } t, \qquad t = 1, \ldots, T \\ P_i &= \text{Index set of periods where individual } i \text{ is observed}, \quad i = 1, \ldots, N \end{aligned}$$

where T is the number of periods with *at least one* individual observed and N is the number of individuals observed *at least one* period. Then the (i, t)-combinations in the data set can be characterized by either the index sets $I_1, \ldots, I_T$ or the index sets $P_1, \ldots, P_N$. This general description may cover any type of panel data. A **balanced** panel data set follows when

$$I_t = \{1, \ldots, N\}, \; t = 1, \ldots, T \Leftrightarrow P_i = \{1, \ldots, T\}, \; i = 1, \ldots, N$$

Other useful notation when dealing with unbalanced panel data:

$$d_{it} = \begin{cases} 1 & \text{if individual } i \text{ is observed in period } t \\ 0 & \text{if individual } i \text{ is not observed in period } t \end{cases}, i = 1, \ldots, N, \; t = 1, \ldots, T$$

implying

$$\begin{aligned} N_t = & \sum_{i=1}^{N} d_{it} = \text{ number of individuals observed in period } t, \\ T_i = & \sum_{t=1}^{T} d_{it} = \text{ number of periods where individual } i \text{ is observed.} \end{aligned}$$

Thus, the *individual-specific mean* of the observed y-values for individual i is

$$\bar{y}_{i.} = \frac{\sum_{t=1}^{T} d_{it} y_{it}}{\sum_{t=1}^{T} d_{it}} \equiv \frac{\sum_{t=1}^{T} d_{it} y_{it}}{T_i} \equiv \frac{\sum_{t \in P_i} y_{it}}{T_i}.$$

Similarly, the *period-specific mean* for period t is

$$\bar{y}_{.t} = \frac{\sum_{i=1}^{N} d_{it} y_{it}}{\sum_{i=1}^{N} d_{it}} \equiv \frac{\sum_{i=1}^{N} d_{it} y_{it}}{N_t} \equiv \frac{\sum_{i \in I_t} y_{it}}{N_t}.$$

Lastly, the *global mean* can be expressed as

$$\bar{y} = \frac{\sum_{i=1}^{N} \sum_{t=1}^{T} d_{it} y_{it}}{\sum_{i=1}^{N} \sum_{t=1}^{T} d_{it}} \equiv \frac{\sum_{i=1}^{N} T_i \bar{y}_{i.}}{\sum_{i=1}^{N} T_i} \equiv \frac{\sum_{t=1}^{T} N_t \bar{y}_{.t}}{\sum_{t=1}^{T} N_t}$$

We assume that **there is no endogenous sample selection**.

QUESTIONS

1. Show, by solving a classical OLS problem where the α_is in Eq.(1) are treated as unknown constants, that the Within estimators of β and α_i, $i = 1, \ldots, N$ can be written as

$$\begin{aligned} \widehat{\beta}_W &= W_{XX}^{-1} W_{XY} \\ \widehat{\alpha}_{iW} &= \bar{y}_{i.} - \widehat{\beta}_W \bar{x}_{i.}, \; i = 1, \ldots, N \end{aligned}$$

   where, in general, W_{ZQ} is defined as

$$W_{ZQ} = \sum_{i=1}^{N} \sum_{t \in P_i} (z_{it} - \bar{z}_{i.})(q_{it} - \bar{q}_{i.})$$

   3 points

2. What do you conclude about the impact of the unbalanced nature of the panel data on the within estimator? Do you need to have succesive observations for each individual?

   1 point

3. Let X denotes all observations on x_{it}, i.e., in the present context,

$$X = \{x_{it} : i \in I_t, t = 1, \ldots, T\} \equiv \{x_{it} : t \in P_i, i = 1, \ldots, N\}$$

   Show that $\widehat{\beta}_W$ is an unbiased estimator of β with $\text{var}(\widehat{\beta}_W | X) = \sigma^2 W_{XX}^{-1}$?
   (Hints: Show first that $\text{var}(W_{XU} | X) = \sigma^2 W_{XX}$)

   4 points

4. Consider now Between estimation of the regression model (1). Summing (1) over $t \in P_i$ and dividing by T_i, we get

$$\bar{y}_{i.} = \alpha_i + \beta \bar{x}_{i.} + \bar{u}_{i.} \tag{2}$$

Can you still assume that the error term in (2), i.e., $\bar{u}_{i.}$, is homoskedastic? (Hints: compute $\text{var}(\bar{u}_{i.}|X)$.)

2 points

5. Propose a transformation of (2) to correct for heteroscedasticity?

1 point

6. Consider now the model defined by Eq. (2) in the absence of individual specific heterogeneity, or $\alpha_i = \alpha$ for all i. It becomes

$$\bar{y}_{i.} = \alpha + \beta \bar{x}_{i.} + \bar{u}_{i.} \tag{3}$$

Apply OLS to (3) to get a first version of *between-individual estimator* for β and α and give the corresponding estimators $\widehat{\beta}_{B1}$ and $\widehat{\alpha}_{B1}$.

4 points

7. Consider now the transformed version of (3)

$$\sqrt{T_i}\, \bar{y}_{i.} = \sqrt{T_i}\, \alpha + \beta \sqrt{T_i}\, \bar{x}_{i.} + \sqrt{T_i}\, \bar{u}_{i.} \tag{4}$$

Apply OLS to (4) to get a second version of *between-individual estimator* for β and α and give the corresponding estimators $\widehat{\beta}_{B2}$ and $\widehat{\alpha}_{B2}$.

4 points

8. What lesson do you draw from the comparison of the estimators $(\widehat{\beta}_{B1}, \widehat{\alpha}_{B1})$ and $(\widehat{\beta}_{B2}, \widehat{\alpha}_{B2})$? (Hints: unbalanced versus balanced panel data.)

1 point

Exercise 2. Application. (**20 points**)

Traffic crashes are the leading cause of death for Americans between the ages of 5 and 32. Through various spending policies, the federal government has encouraged states to institute mandatory seat belt laws to reduce the number of fatalities and serious injuries. In this exercise you will investigate how effective these laws are in increasing seat belt use and reducing fatalities. The data file **Seatbelts.csv** contains a panel of data from 50 U.S. states plus the District of Columbia for the years 1983 through 1997.1 A detailed description is given in **Seatbelts-Description.pdf**.[1]

Questions (**Each question, 2 points**):

1. Describe the data and clean them if necessary. (Balanced panel structure is not necessary afterwards.)

2. Using Pooled OLS, estimate the effect of seat belt use on fatalities by regressing *FatalityRate* on *sb_useage*, *speed65*, *speed70*, *ba08*, *drinkage21*, $\ln(income)$, and *age*. Does the estimated regression suggest that increased seat belt use reduces fatalities?

[1] These data were provided by Professor Liran Einav of Stanford University and were used in his paper with Alma Cohen, "The Effects of Mandatory Seat Belt Laws on Driving Behavior and Traffic Fatalities," *The Review of Economics and Statistics*, 2003, 85(4): 828-843.

3. Do the results change when you add state fixed effects? Provide an intuitive explanation for why the results changed.

4. Do the results change when you add time fixed effects plus state fixed effects?

5. Which regression specification – pooled, within with only state fixed effects, and within with time and state fixed effects – is most reliable? Explain why.

6. Using the results in (4), discuss the size of the coefficient on *sb_useage*. Is it large? Small? How many lives would be saved if seat belt use increased from 52% to 90%?

7. Estimate the same model as in question (3) but now considering that the state effects are random. What assumption are you making now? What are the consequences for the values of estimated parameters?

8. Compare the models estimated in questions (3) and (7) using a robust version of Hausman's test. Explain your approach? What do you deduce about an appropriate estimation strategy?

9. There are two ways that mandatory seat belt laws are enforced: "Primary" enforcement means that a police officer can stop a car and ticket the driver if the officer observes an occupant not wearing a seat belt; "secondary" enforcement means that a police officer can write a ticket if an occupant is not wearing a seat belt, but must have another reason to stop the car. In the data set, *primary* is a binary variable for primary enforcement and *secondary* is a binary variable for secondary enforcement. Run a regression of *sb_useage* on *primary*, *secondary*, *speed65*, *speed70*, *ba08*, *drinkage21*, $\ln(income)$, and *age*, including fixed state and time effects in the regression. Does primary enforcement lead to more seat belt use? What about secondary enforcement?

10. In 2000, New Jersey changed from secondary enforcement to primary enforcement. Estimate the number of lives saved per year by making this change.